Ex No: 2

Date: 30/07/25

PRACTICAL 2 - Types of Network Cables

AIM:

To Study of different types of network cables.

a) Understand different types of network cable.

Different type of cables used in networking are:

1. Unshielded Twisted Pair (UTP) Cable
2. Shielded Twisted Pair (STP) Cable
3. Coaxial Cable
4. Fibre Optic Cable.

Cable type : UTP

Category	Maximum Data Transmission
Category 3	10 bps
Category 5	up to 100 Mbps
Category 5e	1 Gbps

Advantages:

* cheaper in cost
* easy to install as they have a smaller overall diameter

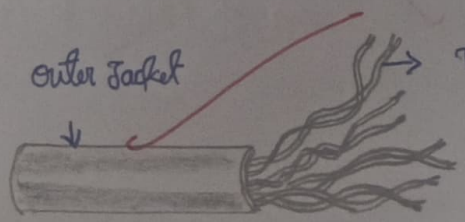
Disadvantages:

* More prone to (EMI) Electromagnetic Interfare and noise

Application / use:

* 10 Base - T Ethernet
* Fast Ethernet, Gigabit Ethernet

Image:



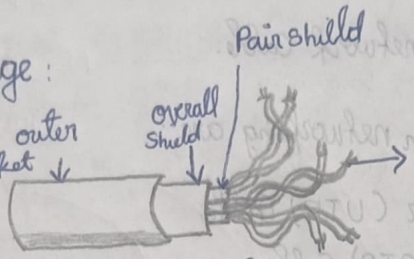
Cable type : STP

Category : Category 6, 6a

Maximum Data Transmission : 10 Gbps

Application/use : Gigabit Ethernet 10 G Ethernet (55m) widely used in data centres.

Image :



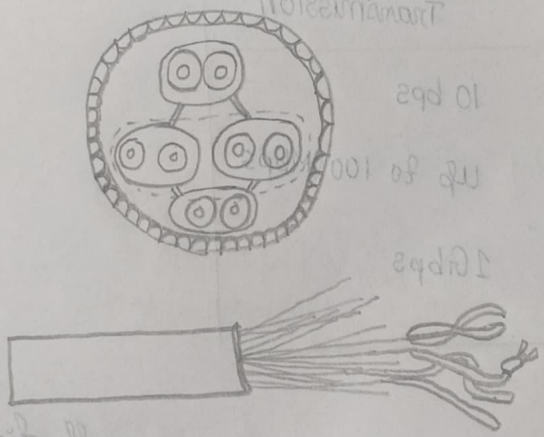
Cable type : SSTP

Category : Category 7

Maximum Data Transmission : 10 Gbps

Application/use : Gigabit Ethernet, 10 G Ethernet (100 m)

Image :



Advantages :

* Shielded
* Faster than UTP
* Less susceptible to noise and interfac

Disadvantages :

* Expensive
* Greater installation effort

Cable Type : Coaxial Cable

Category : RG-6
RG-59
RG-11

Maximum Data Transmission : 10 - 100 Mbps

Advantages :

* High bandwidth
* Immune to interferenc
* ~~Lossloss~~ Low Loss bandwidth
* Versatile

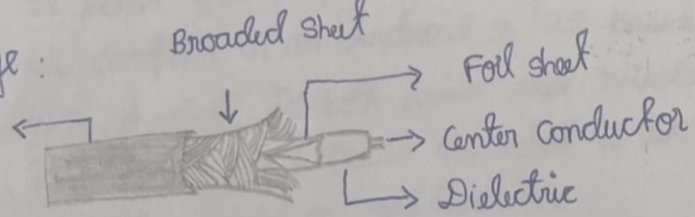
Dis Advantages:

* Limited distanc
* Cost
* Size is bulky

Application/use :

Speed of signal is 500m Television network High Speed internet Connections.

Image :



Cable Type : fibre optics cable

Category : Singl Mode
Mulfi Mode

Maximum Data Transmission : 100 Gbps

Advantages :

* High Speed
* High bandwidth
* High Security
* Long distanc

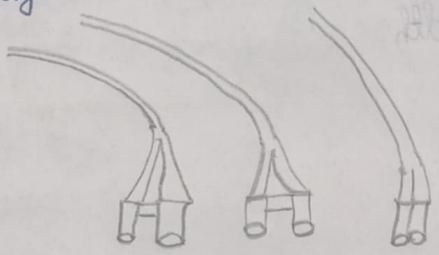
Dis Advantages:

* Expensive
* Required skilled installers

Application/use:

Maximum distance of fibr optics cable is around 100 meters

Image:



RESULT:

Thus studied the different types of network cables